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Title *: **Semi-Decentralized Multi-Spacecraft Collision Avoidance under Communication Constraints**

Short Abstract *

We model spacecraft-to-spacecraft collision avoidance as a semi-decentralized POMDP (SDec-POMDP) where ground-station contacts define synchronization triggers. Two active satellites plan maneuvers under partial observability, sharing information only during visibility windows. Using Recursive Small-Step Semi-Decentralized A* (RSSDA), we compute optimal joint policies balancing miss-distance safety, mission-trajectory deviation, and communication cost. Three scenarios: cooperative shared operator, asymmetric control, costly communication, quantify how contact frequency, symmetry, and cost shape performance vs. centralized and decentralized baselines. Results demonstrate that planned synchronizations achieve near-centralized safety with fewer communications, providing a scalable framework for autonomous CA in crowded LEO.

Topic 1: Space Situational Awareness, Conjunction Analysis, and Collision Avoidance

Topic 2: Machine Learning and Artificial Intelligence Applied to Spaceflight, Space Autonomy

Long Abstract*

Title: Semi-Decentralized Multi-Spacecraft Collision Avoidance under Communication Constraints

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Over the past decade, the number of active satellites has grown rapidly [1], driven in part by the expansion of large-scale commercial constellations. Proposed systems, such as SpaceX’s orbital data center concept, envision deployments of up to one million satellites [2], an increase of approximately two orders of magnitude of current satellite populations. With the growing number of active satellites in LEO, there is a need to understand the dynamics surrounding possible spacecraft-to-spacecraft conjunctions in this crowded environment. Current collision-avoidance (CA) operations are primarily ground-driven and operator-specific: each operator receives Conjunction Data Messages (CDMs) [3] from USSF or commercial SSA providers, where individual operators then evaluate candidate maneuvers and uplink commands according to their own ground contact schedule. However, limited ground visibility, intermittent contact windows, autonomous maneuvering, and multiple operators create scenarios where planning must occur with sparse and outdated state updates rather than continuous information [4]. These constraints naturally motivate decentralized decision-making frameworks, where agents operate with partial information and limited coordination. This exposes a gap in the literature between fully centralized approaches, which assume complete information, and purely local, single-agent methods that neglect interaction effects. In addition, existing work rarely models communication structure directly in the decision-making process, such as intermittent ground contacts and costly coordination.

To address this gap, we model two active spacecraft in a conjunction scenario as a semi-decentralized partially observable Markov decision process (SDec-POMDP) with explicit communication and synchronization events. A POMDP is a mathematical framework for sequential decision-making under partial observability. An SDec-POMDP extends this to multiple agents within a flexible and general communication framework. Synchronization triggers conditioned on any combination of states, actions, observations, or beliefs specify when agents may share information and take joint actions, bridging fully centralized (always synchronized) and fully decentralized (never synchronized) multi-agent planning. In our formulation, these triggers are defined by connections made to different ground-station contact windows of a predefined network. We solve this SDec-POMDP using Recursive Small-Step Semi-Decentralized A* (RS-SDA*) [5], a heuristic search algorithm that computes optimal joint policies for any given communication structure. We build on this prior work by tailoring the model to spacecraft dynamics and communication constraints.

We now detail the key modeling choices for states, actions, rewards, and communication in our multi-agent spacecraft CA SDec-POMDP. The global state comprises Earth-Centered

Inertial (ECI) position and velocity of both spacecraft, plus communications variables that indicate which ground stations are visible and the status of the most recent communication. Actions are a discrete set of small impulsive maneuvers (primarily along-track) applied at a fixed set of decision epochs before closest approach, along with a no-maneuver option.

The reward balances safety, mission performance, fuel use, and communication cost. Safety is enforced via large penalties for trajectories violating a miss-distance threshold at closest approach. Mission performance penalizes deviation from each spacecraft's nominal trajectory, measured via accumulated tracking error or changes in orbital elements. In costly-communication scenarios, we add penalties for synchronization events to quantify trade-offs against safety and mission objectives.

Ground-station contacts are modeled explicitly as deterministic visibility windows that define when each spacecraft can receive updated state estimates, conjunction data, or coordination messages. Contact patterns vary from dense global networks (6–12 stations) to sparse regional coverage (1–3 stations), producing symmetric or staggered schedules across spacecraft. When one spacecraft completes a contact, we model its operator disseminating updated information through coordination channels (e.g., Space-Track.org, operator websites) for the second operator to access before its next contact. We model this as an idealized direct exchange rather than a full dissemination workflow. These asynchronous information availability patterns map directly to SDec-POMDP synchronization triggers, such as updated joint state available via shared channels or recent ground contacts for both spacecraft. Using the Brahe library [6], we compute accurate ground-station visibility and propagate orbits with SGP4 for baseline motion and a high-fidelity numerical integrator for maneuvered arcs, ensuring precise timing of contacts and relative geometry during CA planning.

To compute semi-decentralized joint policies, we apply the existing Recursive Small-Step Semi-Decentralized A* (RS-SDA*) algorithm. RS-SDA* is an admissible heuristic search algorithm that constructs optimal policies offline by breaking the planning horizon into small agent-specific time steps. Each node represents a partial semi-decentralized joint policy up to a given time and communication mode; edges extend it forward by assigning actions (and communication decisions) over the next step for a subset of agents or local observation histories.

RS-SDA* exploits the SDec-POMDP structure for efficiency. Between synchronization events, it optimizes local policies independently for each agent based on their beliefs, keeping the search tractable. At synchronization triggers, it temporarily reasons in a centralized belief space using shared information to coordinate joint actions, yielding tighter heuristic bounds for future value. A*-style pruning discards suboptimal partial policies using a recursive, admissible heuristic. This back-and-forth between decentralized and centralized reasoning provides both scalability and optimality guarantees, making RSSDA well-suited to intermittent communication

patterns like ground-station contacts.

We study three communication scenarios, comparing semi-decentralized policies against centralized planning with perfect information and independent planning with no information synchronization. Metrics include minimum miss distance, cumulative trajectory deviation, total synchronization events, and their timing relative to ground windows and conjunction geometry. (1) In the cooperative shared operator case, both spacecraft are controlled by one operator. We vary ground network density (global vs regional), measuring how contact frequency affects required synchronizations, miss distance, and mission deviation. (2) For the asymmetric control case, we set one spacecraft to be controllable while the other follows a fixed policy (third-party operator). We model a spectrum of third-party behaviors, uncooperative (never maneuvers) and reactive (maneuvers only after its next ground contact if predicted miss distance exceeds its unknown risk threshold). The controllable spacecraft plans conservatively under uncertainty about both the other spacecraft's policy and maneuver timing. (3) Finally, for the last scenario of costly communication, we set synchronization events to incur reward penalties. This specifically helps us quantify the optimal communication budget that minimizes miss distance and trajectory deviation.

In summary, we demonstrate how SDec-POMDPs and RS-SDA* enable modeling of realistic spacecraft-to-spacecraft collision avoidance under intermittent ground-station communication constraints. Our three communication scenarios: cooperative shared operator, asymmetric control, and costly synchronization, quantify how contact frequency, symmetry, and communication cost shape the achievable trade-offs between minimum miss distance, mission-trajectory deviation, and coordination overhead. By formulating multi-spacecraft CA as an SDec-POMDP with ground-network visibility as synchronization triggers, we explore whether planned communication events can achieve near-centralized performance using fewer synchronizations. Our approach naturally extends to probability-of-collision risk models, chance-constrained formulations, larger constellations, and operational integration with existing conjunction assessment pipelines. Results will clarify the minimum communication required for safe autonomous CA in crowded LEO, informing both operator practices and future space traffic management protocols.

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